INSTITUTE OF SPACE TECHNOLOGY
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APP DEVELOPMENT

ASSIGNMENT NO:01

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INTRODUCTION:

Artificial Intelligence (AI) enables computer systems to perform tasks that normally require human intelligence, such as recognizing objects in images and making predictions. In mobile applications, AI can help users understand and process information directly through their smartphones.

EcoSort AI is a smart waste classification Android application. It is designed to help users identify the type of waste by selecting an image from the phone gallery or capturing a new photo with the device camera. After an image is provided, the application runs its classification process and displays a predicted waste category with a confidence score.

The purpose of this project is to demonstrate how an Android application can use an AI model to support waste sorting and encourage better waste management practices.

Objectives:

The main objectives of this assignment are:

• To understand how AI-based features can be used in Android applications.

• To develop and run the EcoSort AI application on an Android device.

• To provide options for selecting an image from the gallery or capturing a photo.

• To classify a waste image and show the predicted category.

• To display the model's confidence score with the prediction.

• To test the main screens and document the application using screenshots.

1. Tools and Technologies

The following tools and technologies are relevant to the development and execution of the EcoSort AI Android application:

Tool / Technology	Purpose
Android Studio	Used as the development environment for building and running the Android application.
Java / Android APIs	Used for application logic and Android device features, depending on the project implementation.
AI Model	Used to analyze a waste image and produce a predicted category and confidence score.
Android Device / Emulator	Used to install, run, and test the application.
Gallery and Camera	Allow the user to provide a waste image for classification.

Application Analysis

EcoSort AI provides a simple interface for waste image classification. The splash screen introduces the application, while the login screen allows the user to access the app. The main classification interface provides the image input area, gallery and camera options, and a result area for the predicted waste category and confidence score.

The supplied application screenshots show the following main features:

• A splash screen displaying the EcoSort AI name and the subtitle “Smart Waste Classification”.

• A login screen with email and password fields and a Login button.

• A main screen where a user can select or capture a waste image and view the classification result.

Development Environment Setup

Android Studio is used as the development environment for the application. The project can be opened in Android Studio, synchronized with Gradle, and then run on a compatible Android device or emulator.

The general steps to run the application are:

• Open Android Studio.

• Open the EcoSort AI Android project.

• Wait for Gradle Sync to complete.

• Check that the required project dependencies and resources are available.

• Connect an Android device or start an emulator.

• Select the target device and click the Run button.

• Confirm that the application opens successfully.

1.1 Splash Screen

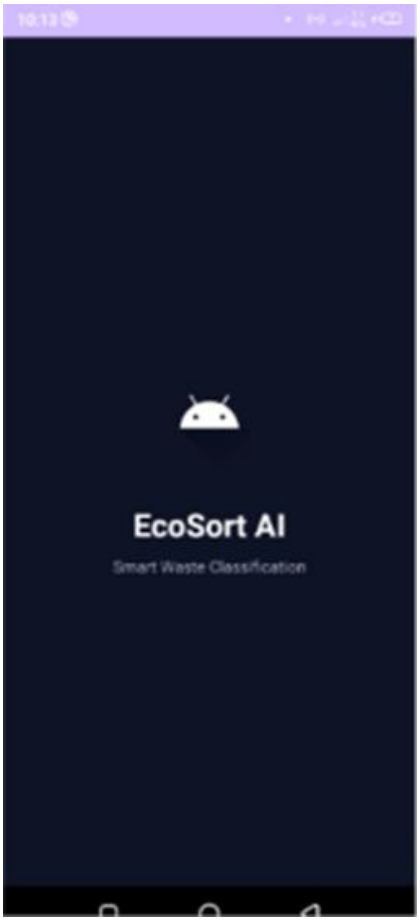


Figure 5.1: EcoSort AI splash screen shown when the application starts.

1.2 Login Screen

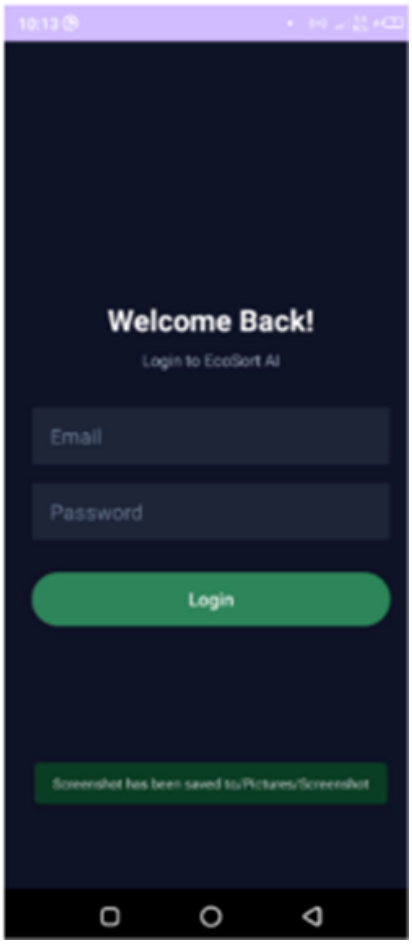
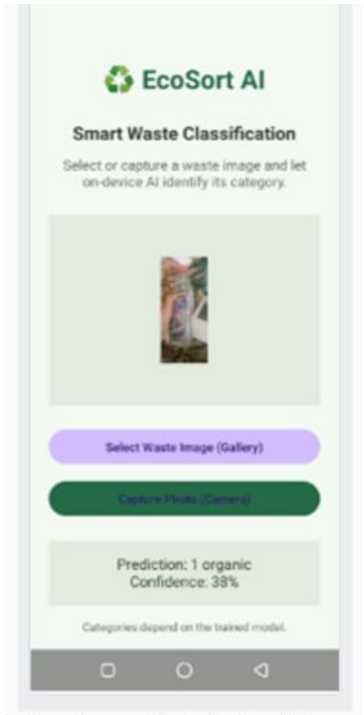


Figure 5.2: Login screen containing email and password fields and a Login button.

Implementation and Main UI

The main interface of EcoSort AI provides options for selecting waste images from the phone gallery or taking real-time photos using the device camera. Once an image is loaded, the user can start the classification process. The AI model analyzes the image and the application displays the predicted waste category along with a confidence score.



ication Screen displaying input image, (38%).

Figure 6.1: EcoSort AI smart waste classification screen showing an input image, prediction (Organic), and confidence score (38%).

The displayed example shows the prediction “Organic” with a confidence score of 38%. The screen also indicates that the available categories depend on the trained model.

AI Model Integration

EcoSort AI uses an AI-based classification process to analyze the waste image provided by the user. The application prepares the selected or captured image and passes it to the integrated model for prediction. The result is then interpreted by the application and shown on the screen.

The general classification process is:

1. The user selects an image from the gallery or captures a photo using the camera.

2. The application loads the image for processing.

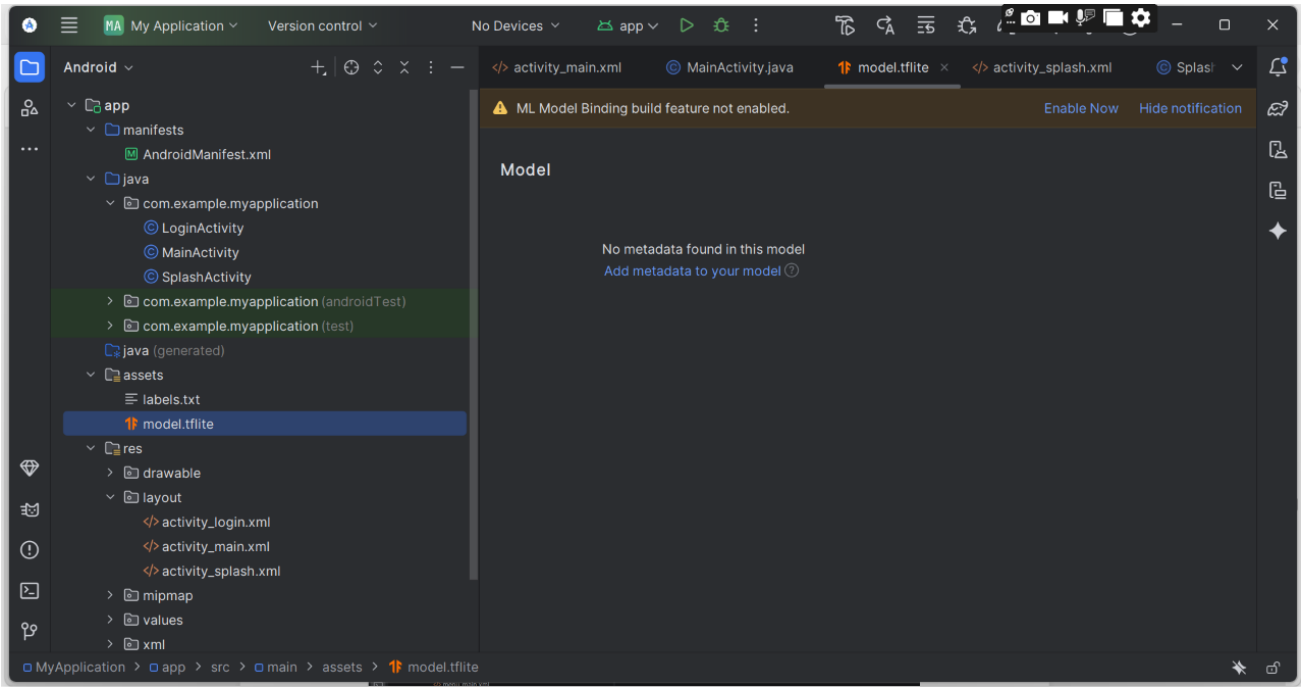
3. The AI model analyzes the image and predicts its waste category.

4. The application receives and interprets the model output.

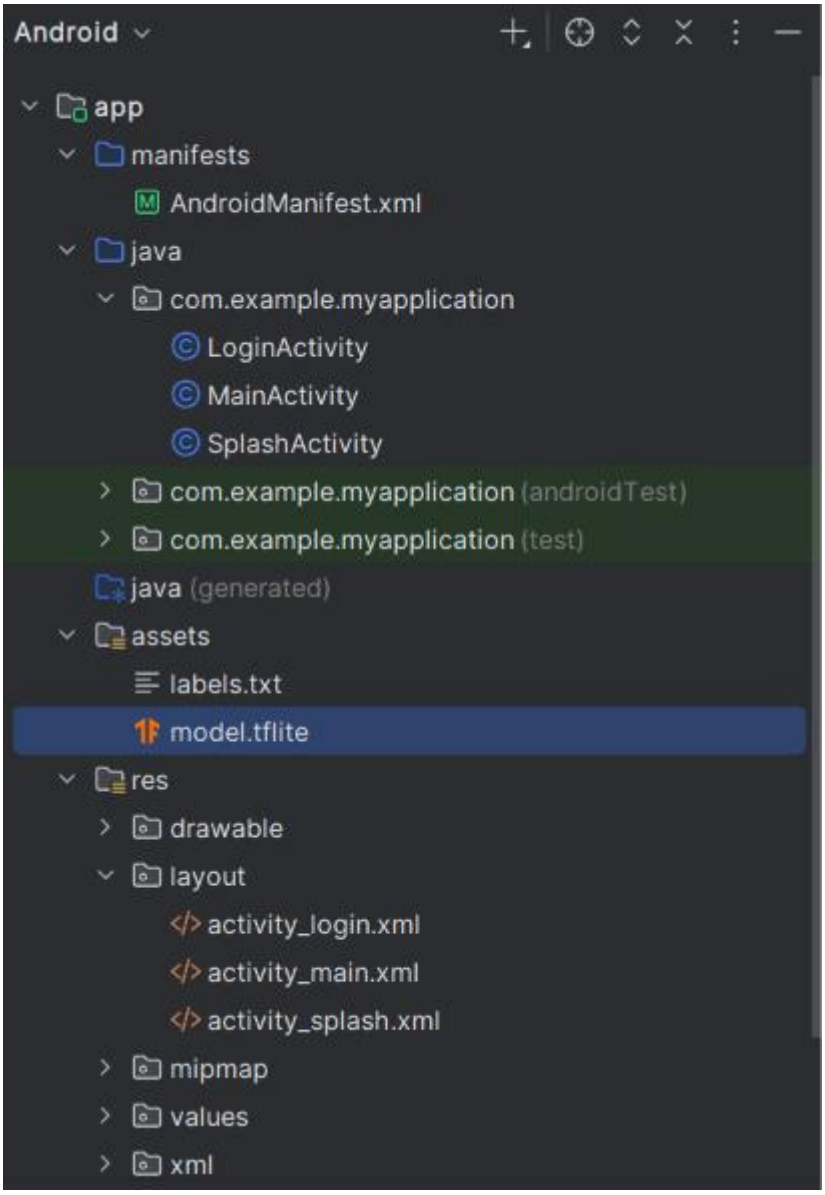
5. The predicted category and confidence score are displayed to the user.

The supplied screenshot confirms that the application displays a category prediction and confidence value. Details such as the model file name, input dimensions, and exact output calculation are not visible in the provided screenshots, so they are not specified in this report.

Digit_model.tflite Inside the Assets



Android Studio Project Structure:



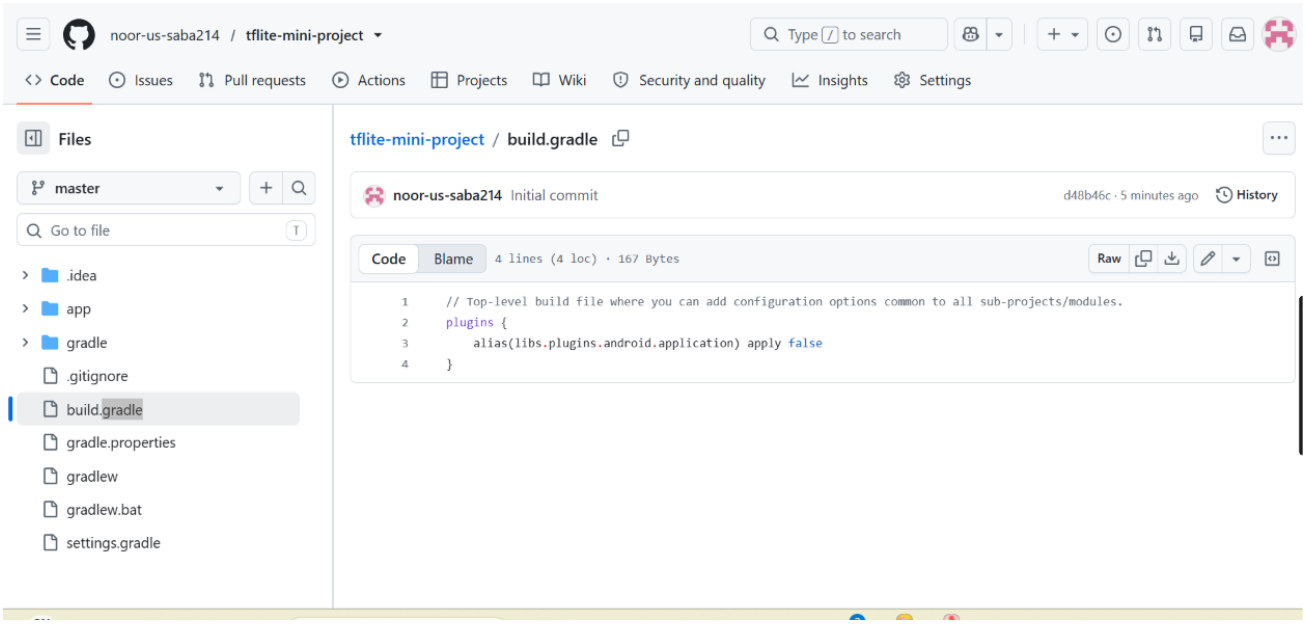
Testing and Results

The application screenshots were reviewed to check the visible behavior of the main screens. The available evidence confirms that the splash screen and login screen are displayed and that the main interface shows an image with a classification result.

Test Case	Action / Input	Expected Result	Observed Result

1	Launch the application	Splash screen should appear	EcoSort AI splash screen is shown
2	Open the login screen	Email and password fields should appear	Login screen is shown
3	View the classification screen	Image input and prediction should be visible	Prediction: Organic; confidence: 38%

GitHub Repository and Uploaded Source Code:



Challenges and Solutions

During the development and testing of an Android application, some common issues may occur. The following points describe practical checks relevant to running and using EcoSort AI:

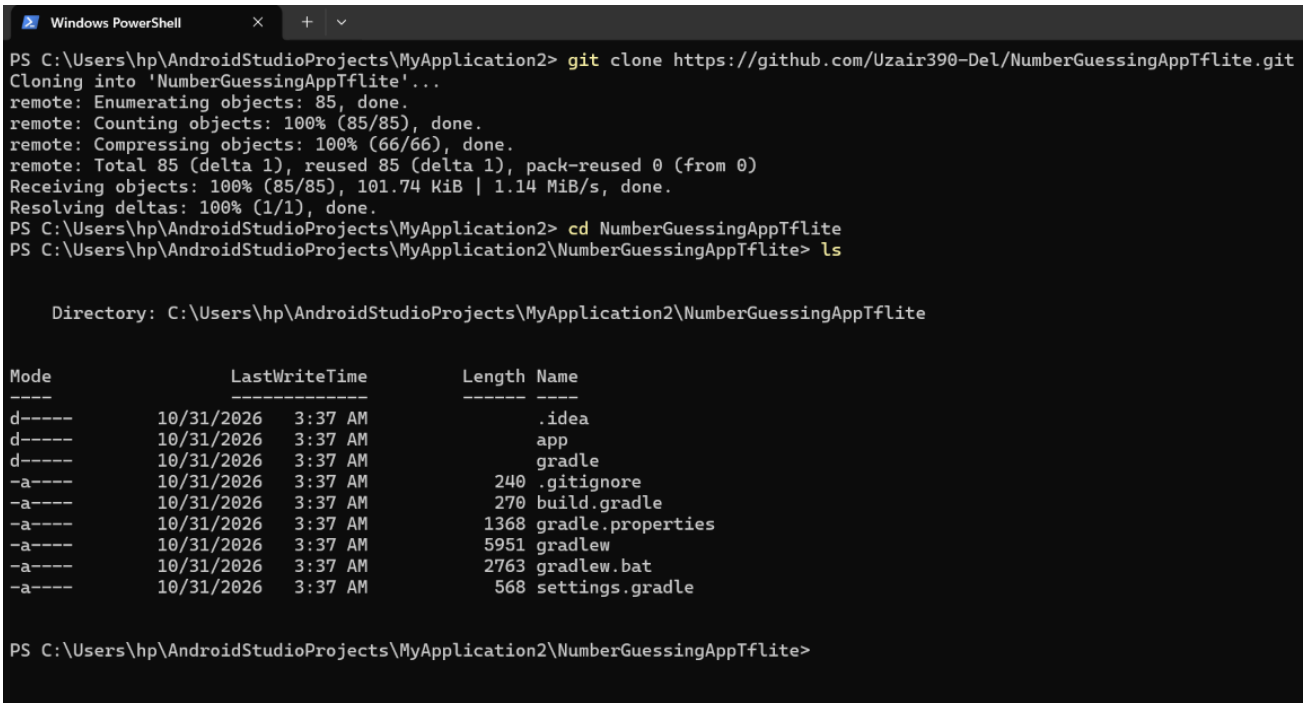
- **Application does not start:** Check the Gradle Sync result, build errors, and device compatibility before running the project again.
- **Gallery or camera image is not available:** Check the required Android permissions and confirm that the image selection or camera action is completed.

- **Image is not classified:** Confirm that a valid image has been selected and that the model resources required by the application are available.

- **Prediction confidence is low:** Try a clearer image with the waste item visible and avoid blurry or poorly lit photos.

Git Command:

git clone <repository-link>



Conclusion

EcoSort AI demonstrates how an Android application can use an AI-based model to classify waste images. The application provides a splash screen and login interface, and its main screen allows users to select or capture a waste image and view a predicted category with a confidence score.

This assignment provides practical understanding of Android application screens, image input through a mobile device, and the use of AI-based prediction in a real-world application. EcoSort AI can support awareness about waste sorting by helping users identify waste categories.

References:

1. Android Developers Documentation: https://developer.android.com/

2. TensorFlow Lite Documentation (for on-device machine learning concepts): https://www.tensorflow.org/lite

3. Screenshots of the EcoSort AI application supplied for this assignment.